

Pallavi Nile

Boston, MA | 617-431-8094 | npallavi0401@gmail.com | [LinkedIn](#) | github.com/pallavi-nile-98 | pallavi-nile-98.github.io

Professional Summary

Software engineer with 18 months of co-op and internship experience in React, TypeScript, Node.js, and AWS. Removed a 24-images/day API cap at The Geode Foundation by serving 1,220 pre-generated images through a deterministic manifest. Seeking a full-time software engineering, cloud, or AI engineering role building on AWS from January 2027.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, Bash

Frontend: React.js, Redux Toolkit, Responsive Web Design

Backend and Data: Spring Boot, JPA, Node.js, Express.js, FastAPI, REST APIs, JWT, PostgreSQL, MongoDB, OpenSearch

Cloud and DevOps: AWS (ECS Fargate, EC2, RDS, S3, ALB, VPC, IAM, CloudWatch), Terraform, Docker, GitHub Actions, Linux

AI and ML: Generative AI, LLM Applications, RAG, Embeddings, Hybrid Search (BM25 + Vector), OpenAI API

Testing and Delivery: JUnit 5, Mockito, Testcontainers, pytest, Playwright, CI/CD, Code Reviews, Agile/Scrum

Professional Experience

The Geode Foundation

Jan 2026 - Jun 2026

Software Engineering Co-op

United States (Remote)

- Eliminated all runtime image-API calls by replacing per-screen AI generation with a deterministic, manifest-driven lookup.
- Cut bundled image metadata 89% (1.7 MB → 189 KB) by building a lean runtime manifest with exact-match and fallback lookup.
- Made shared marketplace links reproduce users' original searches by persisting keywords through React/TypeScript state.

Sumago Infotech Pvt. Ltd.

Jul 2023 - Oct 2023

AWS DevOps Engineer Intern

India

- Shipped MERN applications to AWS (EC2, S3) including Node.js/Express REST APIs with JWT authentication.
- Automated testing and deployment by containerizing applications with Docker and building GitHub Actions CI/CD pipelines.
- Made infrastructure reproducible and observable by writing Terraform templates and CloudWatch monitoring dashboards.

Sumago Infotech Pvt. Ltd.

Sep 2021 - Apr 2022

Full-Stack Development Intern

India

- Improved MongoDB query performance by adding targeted indexes to application queries in a React/Node.js stack.
- Built responsive React.js web applications with reusable frontend components shared across multiple pages.
- Delivered Node.js and MongoDB features in Agile sprints through standups, sprint planning, and Git-based collaboration.

Selected Projects

Computer-Use Capability Engine [GitHub](#)

Python, Playwright, FastAPI, Pydantic, OpenAI API

- Removed the LLM from the execution loop by compiling a once-discovered legacy-UI workflow into a typed, versioned artifact.
- Enforced safe execution with host/action allowlists, approval gates, redacted evidence logs, and same-session human handoff.
- Made every run auditable by validating retries, recoverable states, and hard failures through controlled failure injection.

Claims Approval API [GitHub](#)

Java 21, Spring Boot 3, PostgreSQL, Terraform, AWS (ECS Fargate, RDS, ALB), Docker, GitHub Actions

- Enforced a 4-state claims workflow with separation of duties and optimistic locking in a Spring Boot 3 / Java 21 REST API.
- Covered every state transition and error path with 61 tests (JUnit 5, Mockito, Testcontainers), run in CI in under 1 minute.
- Deployed 39 Terraform-managed AWS resources at ~\$0.07/hour, then induced and resolved 5 failures, documented in a runbook.

AI Research Paper Curator [GitHub](#)

In Progress

Python, PostgreSQL, OpenSearch, Docker, GitHub Actions

- Preserved section context for citations by building async arXiv ingestion with OCR fallback and section-aware chunking.
- Enabled hybrid BM25 + vector retrieval with zero-downtime re-indexing via versioned OpenSearch indices and alias swaps.
- Made pipeline re-runs duplicate-free with an 11-table PostgreSQL schema using unique constraints, backed by 358 CI-run tests.

Education

Northeastern University

Sep 2024 - Dec 2026

M.S., Information Systems

Amrutvahini College of Engineering (AVCOE)

Jul 2019 - Jul 2023

B.E., Computer Engineering

Certification

- AWS Certified Solutions Architect - Associate (SAA-C03)

Mar 2026